

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of M. Tech. (EC)-Examination

December 2012

EC 705 Advance Digital Satellite Communications

Date: 19.12.2012, Wednesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

Section I

- Q 1 a What is the need of satellite communication? 2
b Define following terms 5
ascending node, inclination, apogee, polarization, prograde orbit
c Give the difference between terrestrial communication system and satellite communication. 4
- Q 2 a Write short note Receive-Only Home TV systems. 4
b Explain what is meant by satellite attitude and brief describe two forms of attitude control. 5
c Write technical notes on LEO, MEO, HEO and GEO satellites. 3
- OR**
- Q 2 a Describe telemetry, tracking, and command facilities of a satellite communication system. 6
b Explain layers of atmosphere in detail. 6
- Q 3 a A satellite link operating at 14 GHz has receiver feeder losses of 1.5db and a free space loss of 207 db. The atmospheric absorption loss is 0.5db and the antenna pointing loss is 0.5db. Depolarization losses may be neglected. Calculate the total link loss for clear sky conditions. 2
b An antenna has a noise temperature of 35K, and is matched into a receiver which has a noise temperature of 100K. Calculate (a) the noise power density and (b) the noise power for a bandwidth of 36MHz. 4
c For the system shown in following figures (Fig 1, Fig 2), the receiver noise figure is 12 db, the cable loss is 5 db, the LNA gain is 50 db, and its noise temperature 100K. The antenna noise temperature is 35 K. Calculate the noise temperature referred to the input. 6

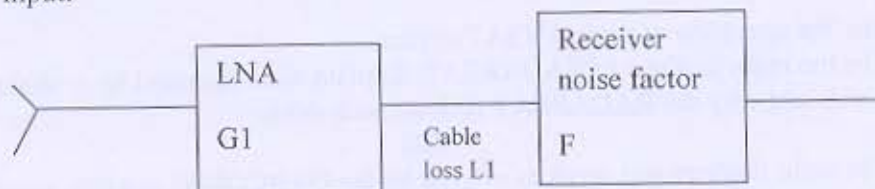


Fig 1

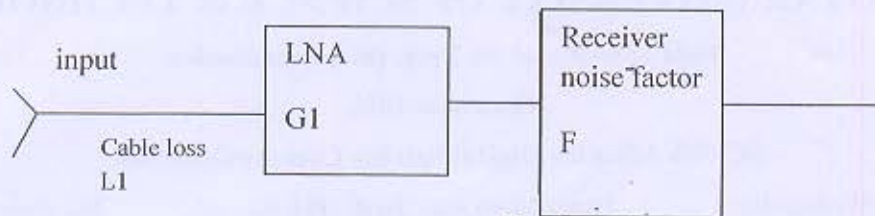


Fig 2

OR

- Q 3**
- a Calculate rain attenuation for frequency of 12GHz (for horizontal and vertical polarization and circular polarization), the rain attenuation which is exceeded for 0.01% of the time and in any year, for a point rain rate of 10mm/h. The earth station altitude is 600m, and the antenna elevation angle is 50 degrees. The rain height is 3 km. ($a_h=0.0188$, $a_v=0.168$, $b_h=1.217$, $b_v=1.2$ for $f=12\text{GHz}$) 6
 - b Determine the rate of regression of nodes and rate of rotation of lines of apside for $I=98.6328$ deg, $e=0.0011501$, $n=14.23304826$ day⁻¹, $a=7192.3$ km. 4
 - c A satellite is orbiting in the equatorial plane with a period from perigee to perigee of 12hr. Given that the eccentricity is 0.002. Calculate the semi major axis. The earth equatorial radius is 6378.1414km. 2

Section II

- Q 4**
- a Explain What is meant by single entry interference. 1
 - b Explain What is meant by energy dispersal. 1
 - c Explain the cross polarization discrimination and polarization isolation 2
 - d Explain how depolarization is caused by rain. 2
 - e What is unique word? 1
 - f An FM/TV carrier is specified as having a modulation index of 2.571 and a top modulating frequency of 4.2 MHz. Calculate the protection ratio required to give a quality impairment factor of (a) 4.2 (b) 4.5. 2
 - g What is meant by effective path length in connection with rain attenuation? 2
- Q 5**
- a Explain the possible interference modes satellite circuit and terrestrial station 6
 - b Describe the channeling skin foe SPADE system for Intelsat satellite. 6
- OR
- Q 5**
- a Write technical note on time division multiplexing and discuss various errors associated with it. 6
 - b Write technical note on GPS. 6
- Q 6**
- a Describe the operation of typical VSAT system. 6
 - b Describe the main feathers of RADARSAT. Explain what is meant by a 'down to dusk' orbit and why the RADARSAT follows such orbit. 6
- OR
- Q 6**
- a Describe main feathers and services offered by the ORBCOMM satellite system. 6
 - b Describe the principles of operation code division multiple access. What effect do the unwanted signals have on the wanted signal? 6